```
clc;
clear;
close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)];
n = 0:N-1;

%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x);

% Direct Form-II Transposed realization
y2 = filter(b,a,x);

%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

%% 3. Parallel Realization
[r,p,k] = residuez(b,a);

% Direct polynomial term
yp = filter(k,1,x);

for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end

% Keep real-valued results when imaginary parts are numerical roundoff
y1 = real(y1);
y2 = real(y2);
yc = real(yc);
yp = real(yp);

%% 4. Display Coefficients
disp('Numerator coefficients b =');
```

```
Numerator coefficients b =
```

```
disp(b);
```

```
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
disp('Denominator coefficients a =');
```

```
Denominator coefficients a =
```

```
disp(a);
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
disp('Second Order Sections (Cascade) =');
```

```
Second Order Sections (Cascade) =
```

```
disp(sos);
```

```
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327
```

```
disp('Cascade Gain =');
```

```
Cascade Gain =
```

```
disp(g);
```

```
    0.0048
```

```
disp('Residues (Parallel) =');
```

```
Residues (Parallel) =
```

```
disp(r);
```

```
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i
```

```
disp('Poles (Parallel) =');
```

```
Poles (Parallel) =
```

```
disp(p);
```

```
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i
```

```
disp('Direct term =');
```

```
Direct term =
```

```
disp(k);
```

```
    0.0256
```

```
%% 5. Error and Percentage Deviation
E = [max(abs(y1-y2)); ...
     max(abs(y1-yc)); ...
     max(abs(y1-yp))];

reference = max(abs(y1));

if reference > 0
    D = E/reference*100;
else
    D = zeros(size(E));
end

Structure = {'Direct Form-I'; ...
             'Direct Form-II'; ...
             'Cascade'; ...
             'Parallel'};

Maximum_Error = [0; E];
Deviation_percent = [0; D];

T = table(Structure,Maximum_Error,Deviation_percent);

disp('Comparison Table:');
```

```
Comparison Table:
```

```
disp(T);
```

Structure	Maximum_Error	Deviation_percent
{'Direct Form-I' }	0	0
{'Direct Form-II'}	0	0
{'Cascade' }	1.9221e-15	8.242e-13
{'Parallel' }	1.9811e-15	8.4949e-13

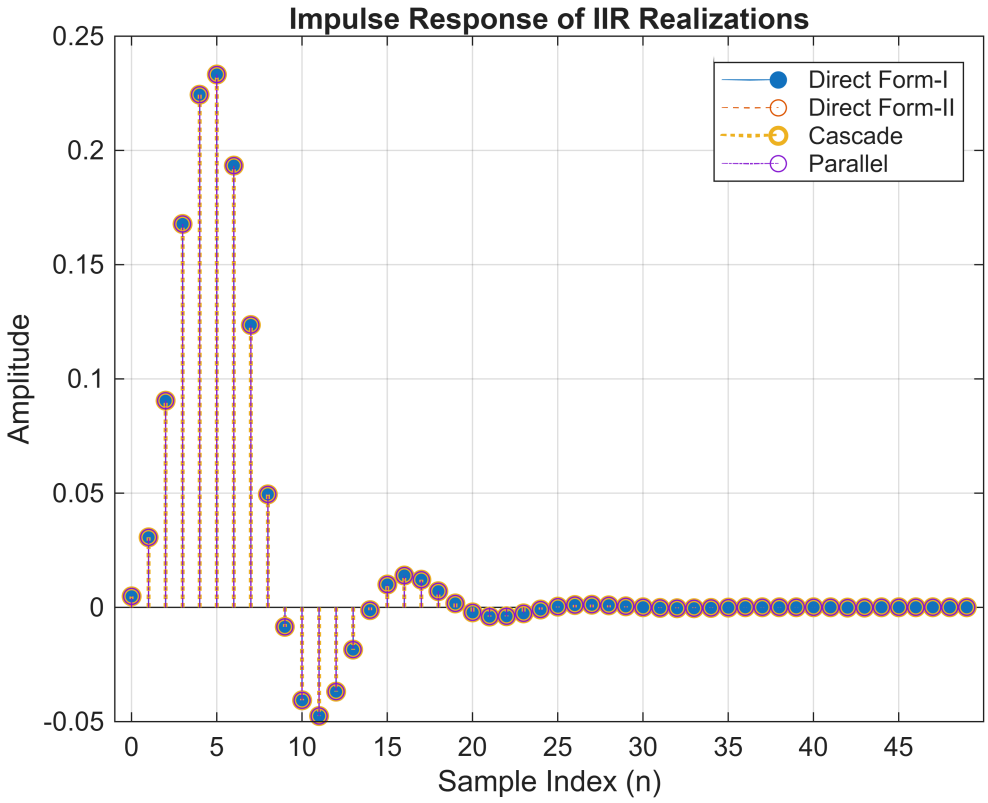
```
%% 6. Impulse Response Comparison
figure('Name','Impulse Response Comparison');

stem(n,y1,'filled');
hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
stem(n,yp,'-.');

grid on;
```

```
xlabel('Sample Index (n)');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');

legend('Direct Form-I','Direct Form-II', ...
       'Cascade','Parallel','Location','best');
```

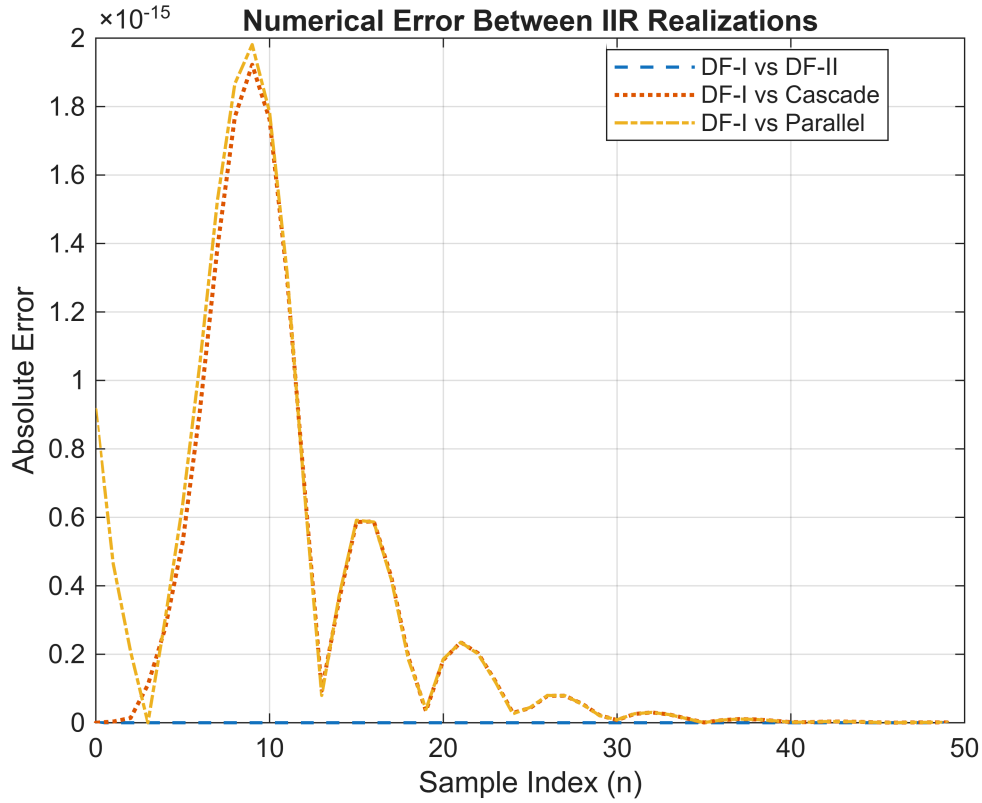



```
%% 7. Error Plot
figure('Name','Numerical Error Comparison');

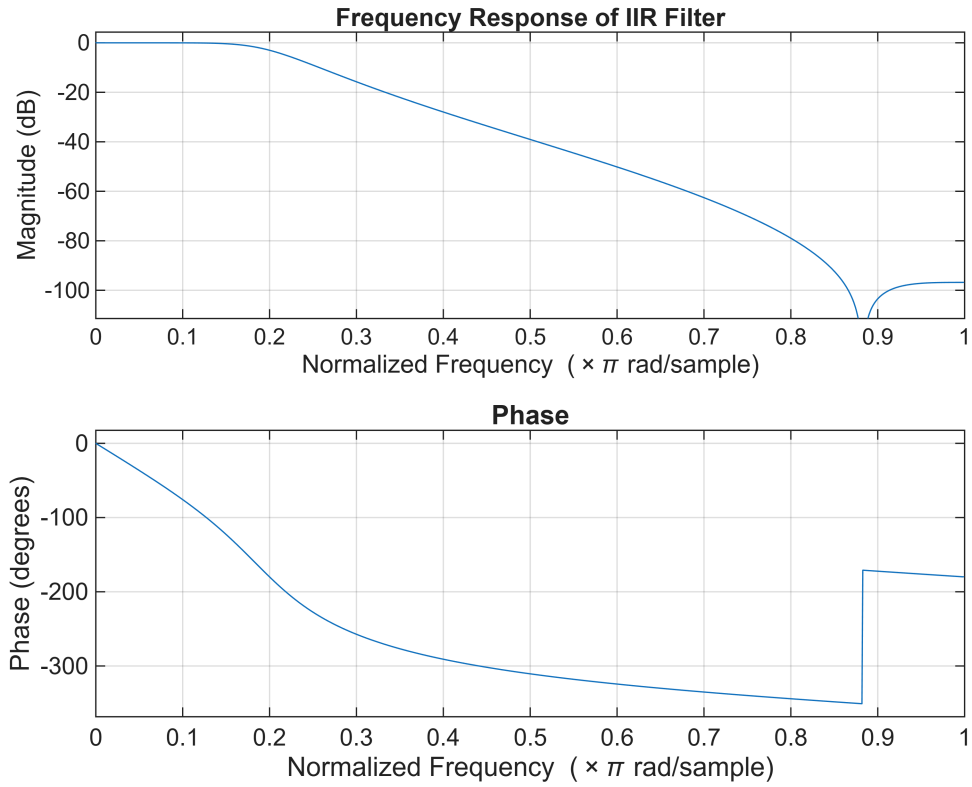
plot(n,abs(y1-y2),'--','LineWidth',1.2);
hold on;
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);

grid on;
xlabel('Sample Index (n)');
ylabel('Absolute Error');
title('Numerical Error Between IIR Realizations');

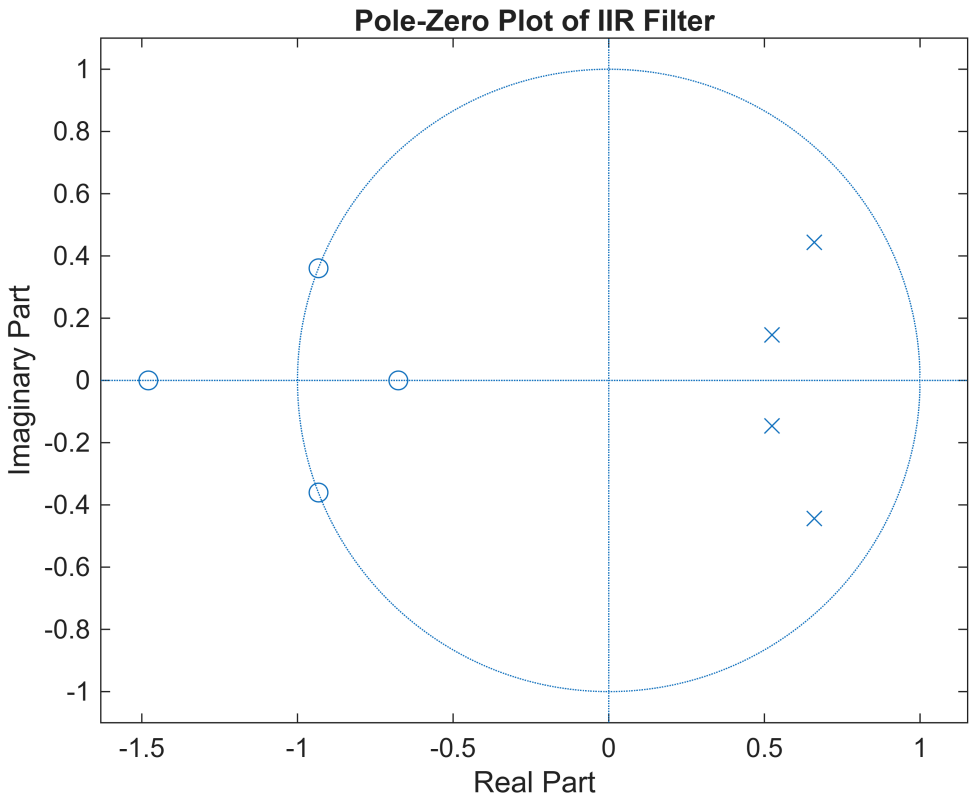
legend('DF-I vs DF-II','DF-I vs Cascade', ...
       'DF-I vs Parallel','Location','best');
```




```
%% 8. Frequency Response
figure('Name','IIR Frequency Response');
freqz(b,a,1024);
title('Frequency Response of IIR Filter');
```




```
%% 9. Pole-Zero Plot
figure('Name','IIR Pole-Zero Plot');
zplane(b,a);
title('Pole-Zero Plot of IIR Filter');
```


Pole-Zero Plot of IIR Filter
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1